Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach repairs as testable proposals rather than automatic fairness guarantees.

Supply all count records, preserve their populations and assess the new unequal-group case independently.

Keep one habit: Require the selection process and remaining limitations alongside every combined percentage.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Identify selection and coverage **bias**

Teach Week 20 as a two-session investigation using fictional data. A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents.

If only club members are surveyed, an 80 percent result misses a group with a different pattern. Examine who is included before treating a larger sample as better.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can distinguish a **bias**ed collection **process** from a mere small dataset.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

The population is the full target group; the sample is the observed subset.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Sampling and Representation

Noise and Outliers

Decision Trees

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

Concept to explain

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.

Term	Meaning
bias	A systematic tendency in data, decisions or outcomes that can distort a result or disadvantage people.
rebalance	Change the relative representation of groups or classes in data using a stated method.
subgroup	A defined subset of cases considered separately in analysis.
</details>

Term	Meaning
retest	Run a test again after a change or under specified new conditions.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach observed combination before a population claim

> Keep the three supplied count records distinct.

State the opening poll counts

A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents. If only club members are surveyed, an 80 percent result misses a group with a different pattern.

Examine who is included before treating a larger sample as better.

Predict and run the complete script

```
club_yes, club_n = 8, 10
other_yes, other_n = 2, 10
print(club_yes / club_n)
print(other_yes / other_n)
print((club_yes + other_yes) / (club_n + other_n))
```

Explain the denominator and scope

The combined observed sample has 50 percent Yes. This is not automatically a population estimate: we also need to know how the groups were selected and how many people each represents.

Do not average percentages without considering their denominators.

Collect the90/10 changed-mix calculation

> **Suppose 90 club respondents give 72 Yes and 10 others give 2 Yes. What is the combined observed proportion?**

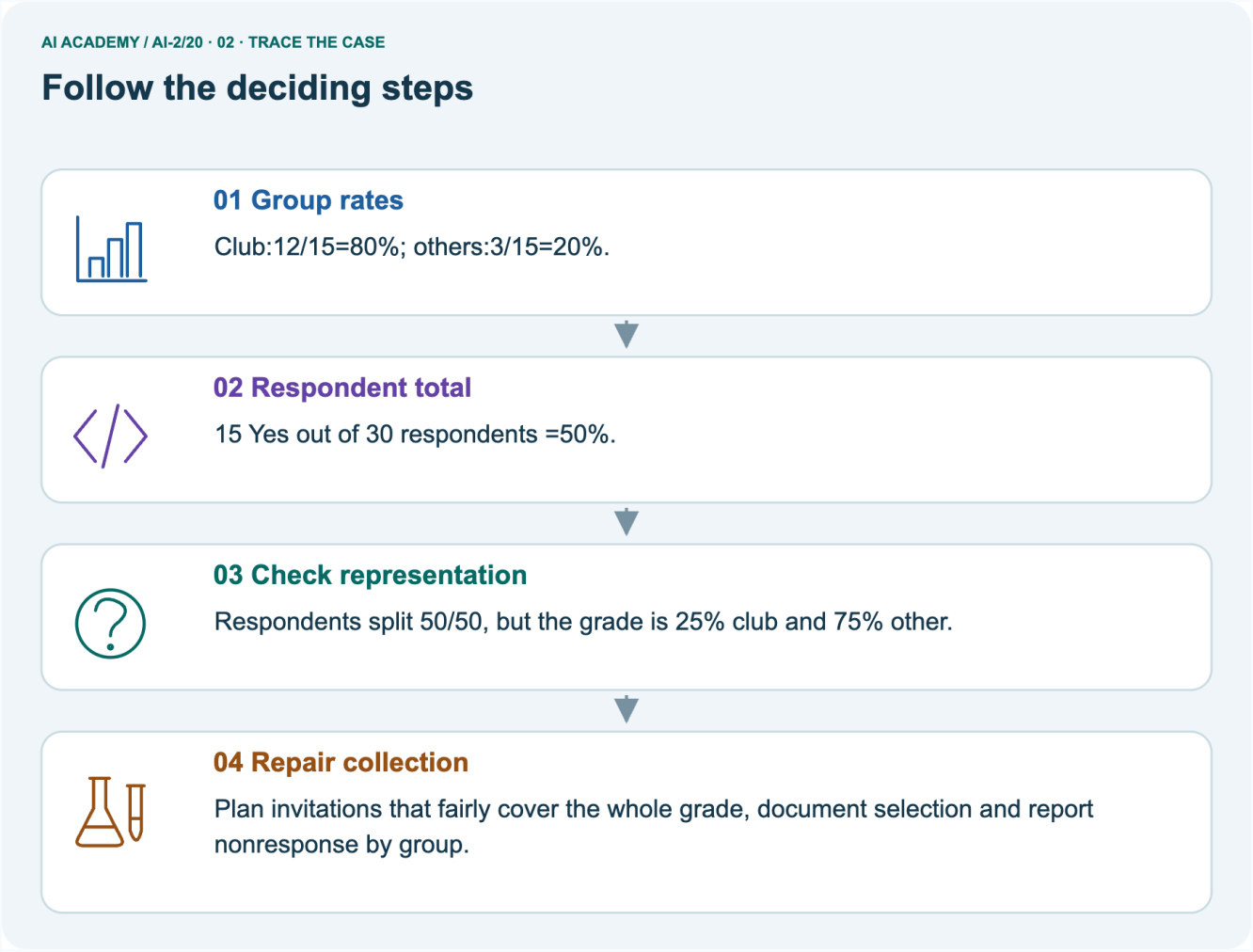
Introduce the separate volunteer and grade counts

A fictional school polls learners about adding comic recommendations.

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

Trace the weighting illustration and its assumptions



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Interpret more club volunteers

Change: Recruit more club volunteers only.

Prediction: Sample size increases without fixing missing representation of other learners.

Reason: The added respondents come from the already overrepresented group.

Elicit an evidence-limited explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Audit: quiet-room voices work; noisy-room voices often fail.

2. Possible causes: missing environments, microphone differences, narrow speakers.

3. Repair: collect consented examples across settings; remove shortcut features.

4. **Retest** the same **subgroup** cases and report before/after results.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Exact explanation to model

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Explain the added worked example

Key terms

Term	Meaning
Selection bias	Systematic distortion caused by who enters the data
Undercoverage	Leaving part of the target group out of selection
Nonresponse	A selected participant does not respond
Subgroup	A defined part of a larger group

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Repair Bias in Data

The facts we will use

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club:12/15=80%; others:3/15=20%.



2. Respondent total

15 Yes out of 30 respondents =50%.



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group rates	Club:12/15=80%; others:3/15=20%.
2. Respondent total	15 Yes out of 30 respondents =50%.
3. Check representation	Respondents split 50/50, but the grade is 25% club and 75% other.
4. Repair collection	Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

Expected reasoning and fading help

Teacher check: 5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

It assumes each volunteer group rate represents its whole group; that assumption has not been established.

Changed-case reasoning: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: **Training** has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Write the question to explain: Propose repair and evaluation.

Reveal after **prediction**s: Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Use the earlier changed question as optional later practice. Reserve Audit independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

74/100 = 74 percent. The sample is dominated by club respondents. Exit: No. Existing interest could influence club membership, and other factors may differ.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Opening group rates80%,20%; combined10/20=50%. Controlled mix72/90 and2/10 keeps those rates but combines74/100=74%. Separate comic volunteers12/15 and3/15 combine15/30=50%; grade shares25%/75% give illustrative35%, not an established population result. W2 combines5/10=50%; averaging75% and33.3% is a different calculation. A collection repair is a proposal until new evidence exists; no model test improvement is supplied. Reserve18/20 and2/10 for independent W4.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied two-scenario respondent-mix audit. Calculate group and combined rates, explain selection limits and propose a collection repair. No retrained model or **subgroup**-test outputs are supplied; do not describe a proposed repair as an observed improvement.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim

Record	What to capture
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: A 6/8 and B 1/2 combine to 7/10 = 70 percent, with group rates 75 and 50 percent. Require a selection limitation.

A convenient shortcut preserves meaning: Surveying another 100 people from the same undercovered source may make the sample larger without fixing the selection problem.

Improve access to missing groups and record **nonresponse**; do not simply duplicate existing rows.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Locate undercoverage

A morning-only survey claims to represent all library visitors. Which visitors may be missed?

Afternoon or evening visitors, and people unable to attend in the morning.

W2 Combine counts

Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

W3 Repair the collection

A form is shared only in the robotics club. How can the **process** better cover the grade?

Use a full-grade selection frame and access channels available across groups, with a **nonresponse** record. State remaining limitations.

W5 Avoid causal claims

Club members favor coding more often. Does club membership necessarily cause that preference?

No. Existing interest could influence club membership, and other factors may differ.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Additional transfer practice question

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

Reasoned practice answer

Inspect representation and recommendation results for poetry readers. Gather relevant varied examples and compare performance for both interests on separate test cases.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Duplicating one poetry record supplies diverse **evidence**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Add varied relevant poetry titles with consistent metadata, keep scoring fixed and compare results on the same validation requests by interest. Then inspect whether scoring still disadvantages poetry.

Save fresh final requests and do not infer cause from suggestion counts alone.

Explanations for the additional workbook examples

W1 Locate undercoverage

W2 Combine counts

W3 Repair the collection

W4 Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade.

W5 Avoid causal claims

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Change feature coverage or weights only after locating which reader profiles are underserved.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use two colors of anonymous group cards. Extension: compare combined rates after changing only sample group sizes while preserving group rates.

Independent W4 **evidence**: 90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade. Record whether the learner needed no help, a hint or explanation.

Fresh retry: A 6/8 and B 1/2 combine to 7/10 = 70 percent, with group rates 75 and 50 percent. Require a selection limitation.

Plan the delayed check

Suggested return points in this level: ai-2/21, ai-2/23, ai-2/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 21: Similarity by Selected Traits. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week21: Similarity by Selected Traits. Carry forward coverage, feature definitions and denominator limits; similarity scores need a declared comparison rather than a claim that a convenient sample represents everyone.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Identify selection and coverage **bias**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

Reasoned practice answer: 5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

Check every learner before discussion: Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Recruit more club volunteers only. Require a **prediction** and a reason before comparing. Expected result: Sample size increases without fixing missing representation of other learners.

Reason: The added respondents come from the already overrepresented group.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

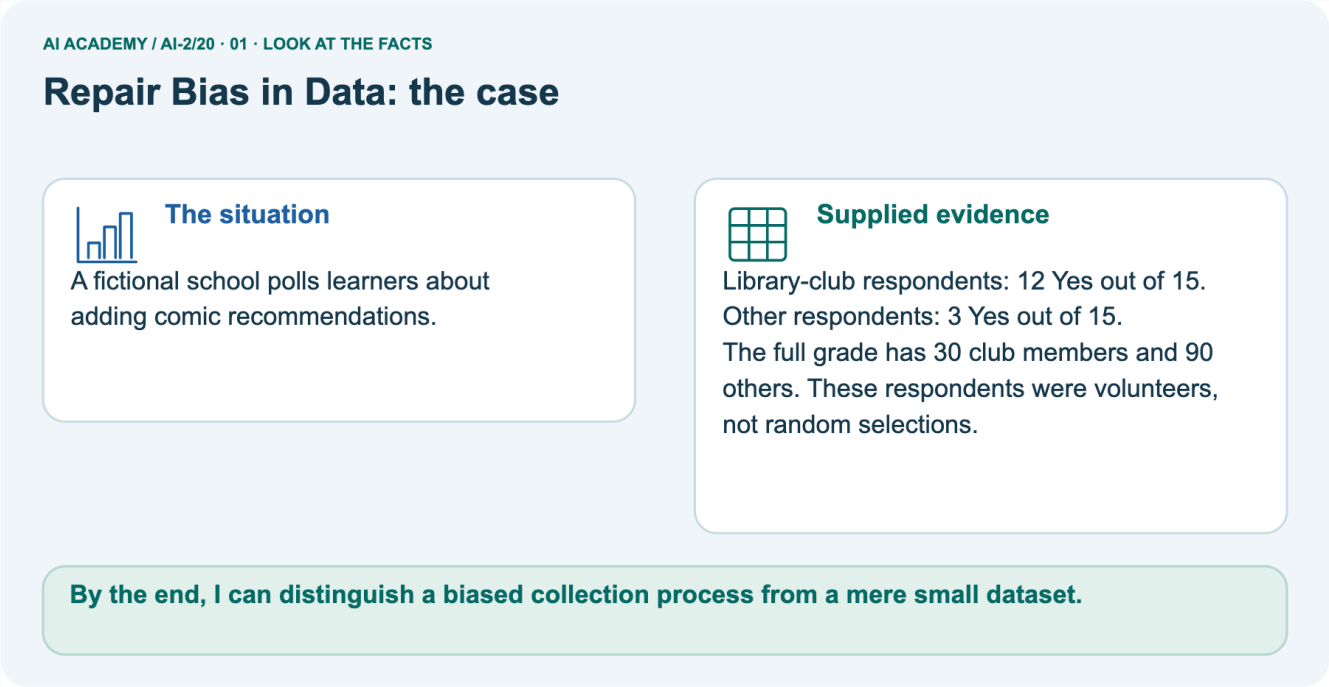
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Repair Bias in Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

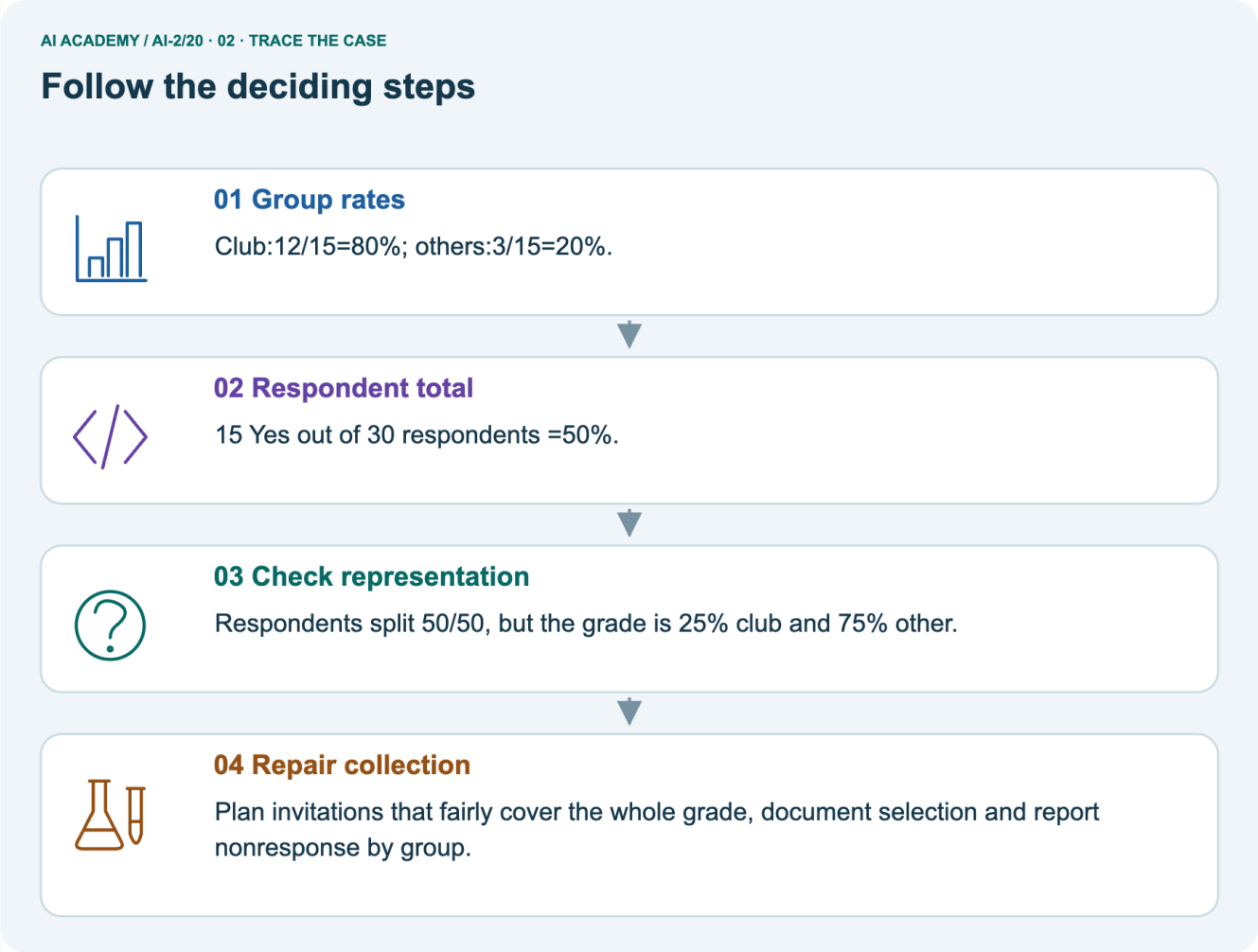
A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

These respondents were volunteers, not random selections.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



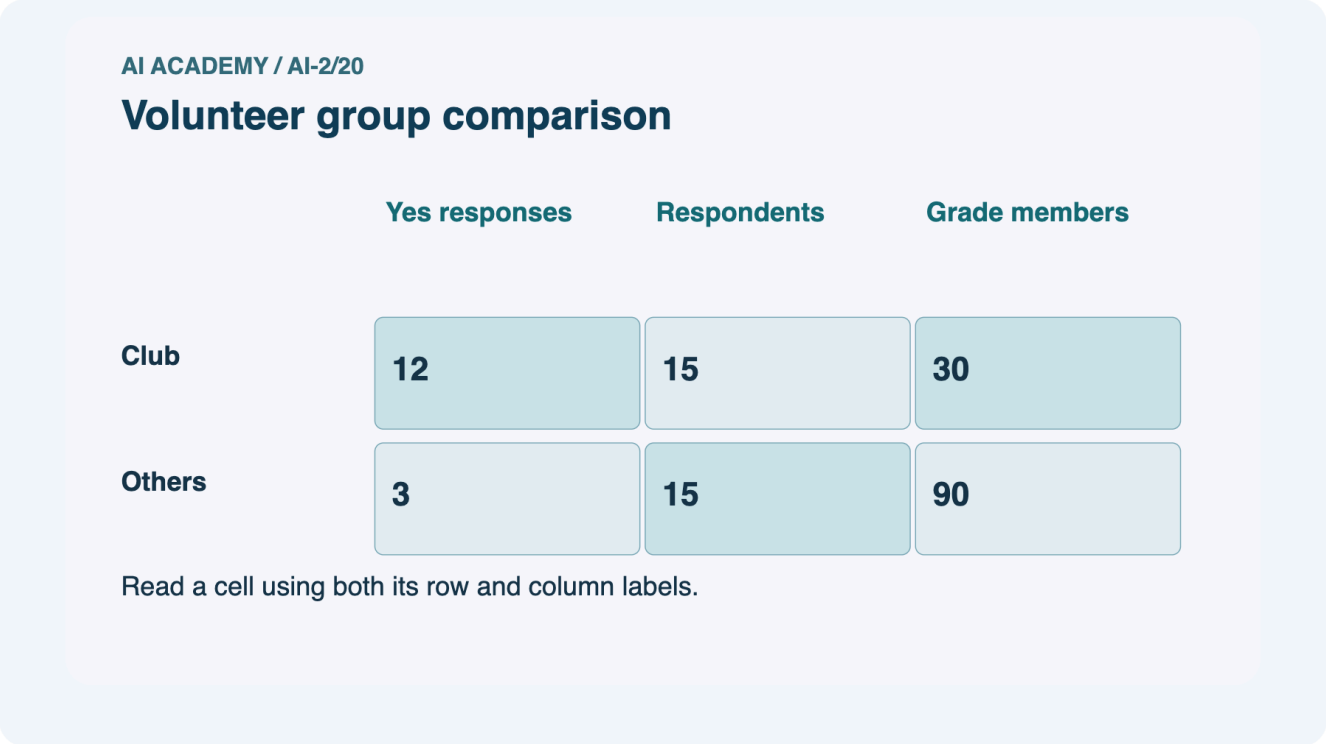
Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Volunteer group comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

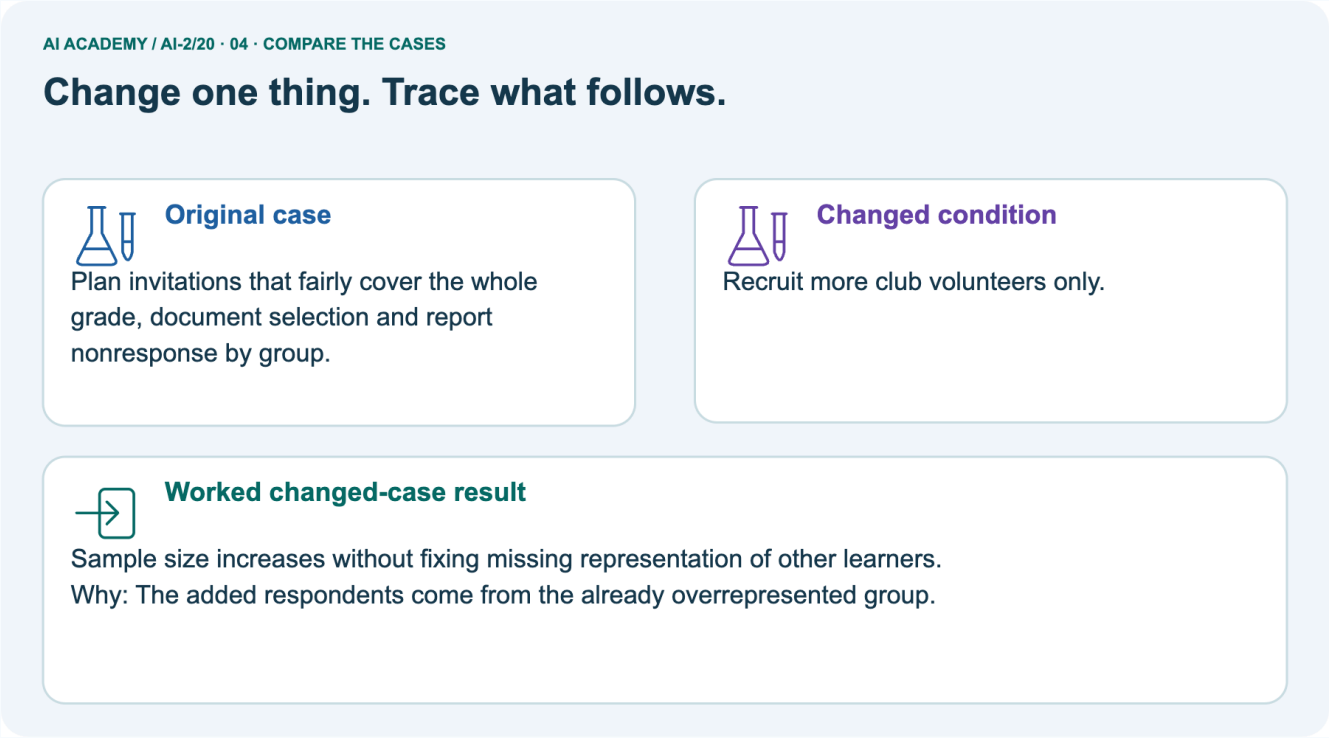
Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group. Changed condition: Recruit more club volunteers only.

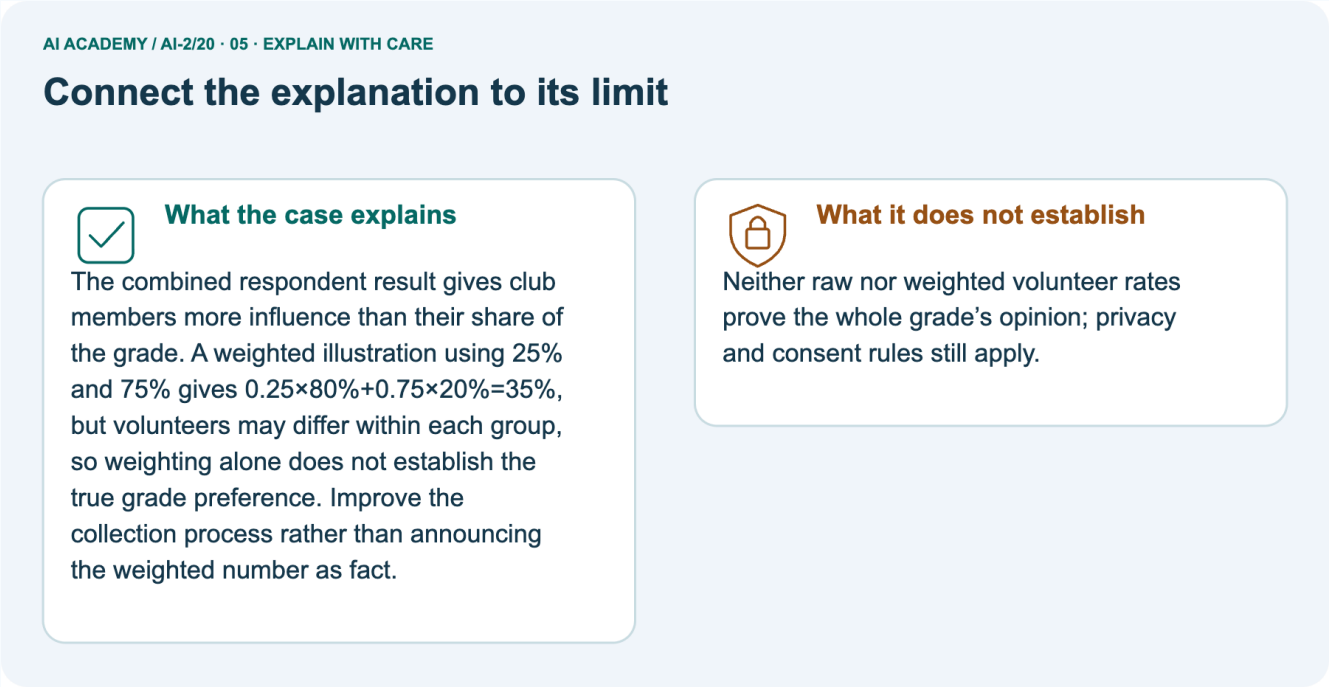
Expected result: Sample size increases without fixing missing representation of other learners. Reason: The added respondents come from the already overrepresented group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The added respondents come from the already overrepresented group.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

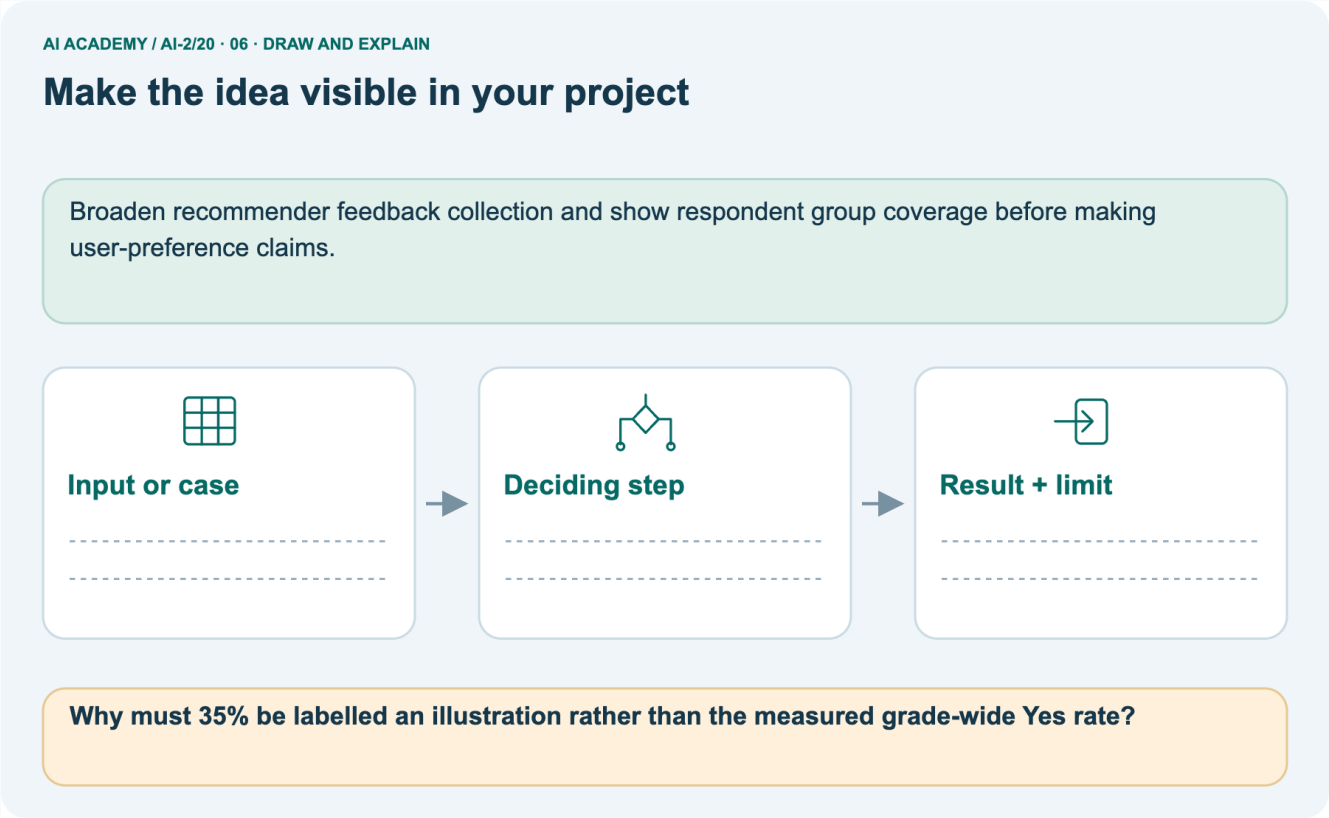
Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It assumes each volunteer group rate represents its whole group; that assumption has not been established.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Repair Bias in Data: the case

02. Follow the deciding steps

03. Volunteer group comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.